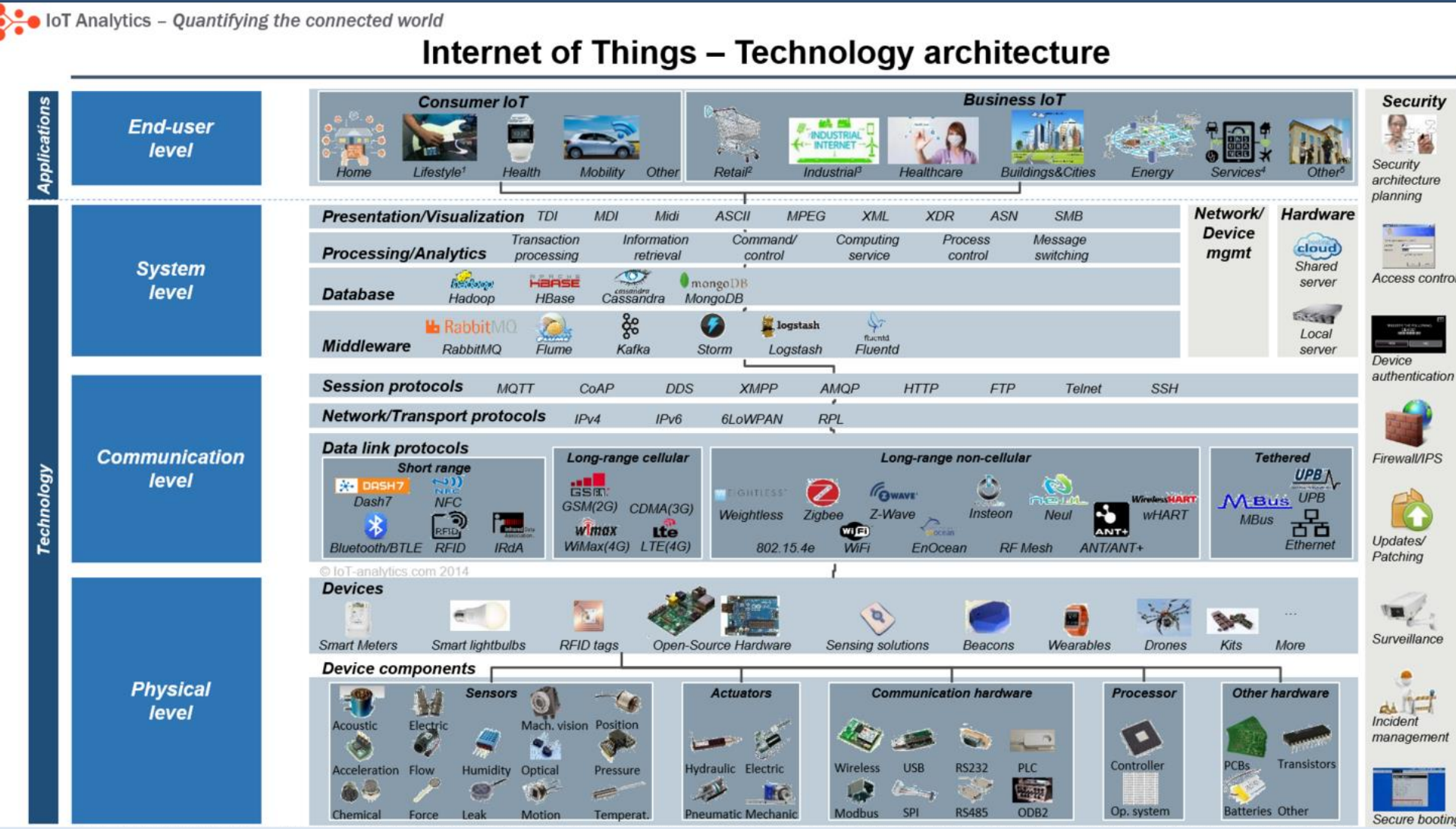
SAKARYA ÜNİVERSİTESİ
Bilgisayar ve Bilişim Bilimleri Fakültesi
Bilgisayar Mühendisliği Bölümü

BSM 451
INTERNET OF THINGS AND APPLICATIONS

IoT Wireless Communication Technologies
Wireless Sensor Networks (WSN)

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Researcher Ünal ÇAVUŞOĞLU

IoT Technology Architecture



IoT Wireless Communication Technologies

- Short Range
  - Radio-Frequency Identification, RFID,
  - Near Field Communications, NFC,
  - Bluetooth Low Energy (BLE),
  - Infrared (IRdA)
- Non-Cellular Long-Range
  - **Wireless Sensor Networks (IEEE 802.15.4 - ZigBee)**,
  - Z-Wave
  - ANT/ANT+
- Cellular Long-Range
  - GSM / GPRS
  - 3G / 4G (LTE) / 4.5G (LTE Advanced)
  - WiMAX

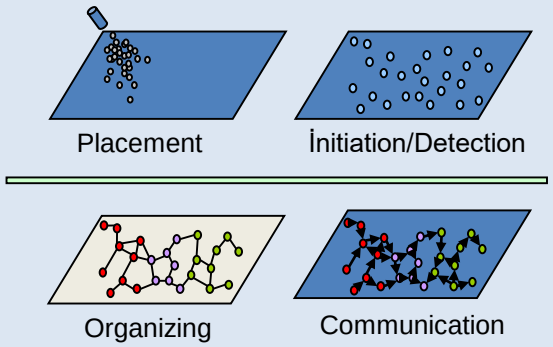
Wireless Sensor Networks (WSN)

- Wireless Sensor Networks are formed by communicating sensor nodes.

- Sensor node
  - Can sense physical attributes in its environment
    - Humidity
    - Temperature
    - Light, etc.
  - Limited processing capability
  - Short distance wireless communication
  - Small size
  - Low power
  - Low cost integrated circuit



- Basic features
  - Random placement
  - Self-organizing
  - Collaboration
  - Local computing

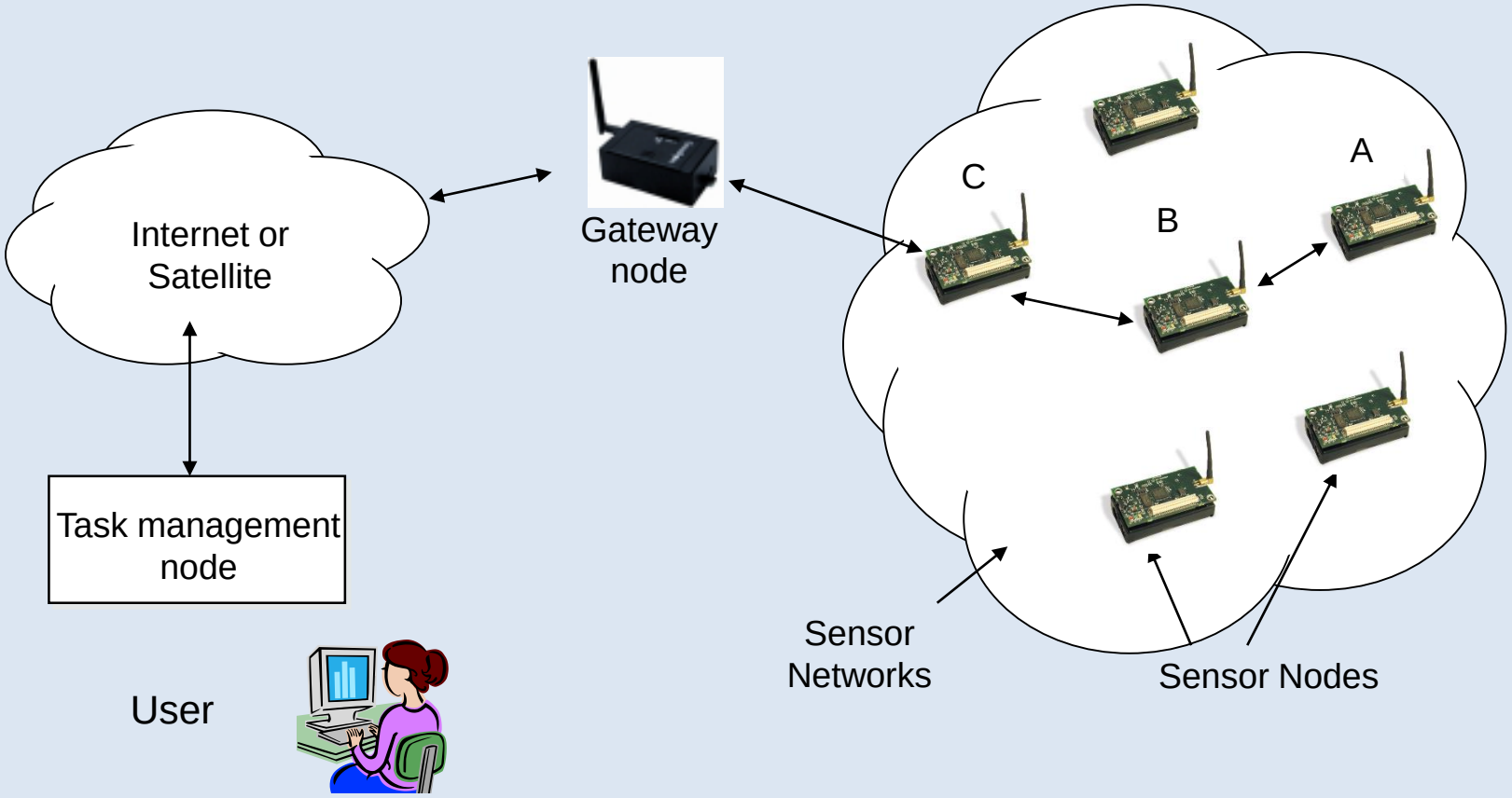


Sensing Technology Examples Used in WSN

- Temperature/heat
- Humidity
- Accelaration
- Pressure
- Light
- Noise level
- Mechanic tension
- Determination of existence or absence of an object
- Speed, size, direction of a moving object
- Electromagnetic field
- Picture
- Voice
- Soil composite



WSN Architecture



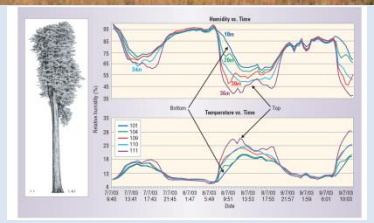
- There may be hundreds, thousands, or even hundreds of thousand node can be in a WSN depending on the application..

WSN Applications

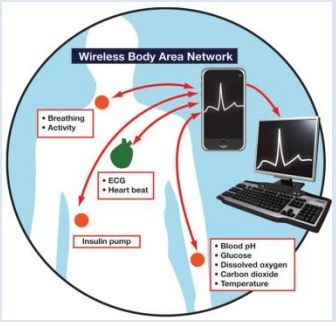
- Military
  - Target detection, attack detection, surveillance of war field,
  - Differentiation of ally/enemy, etc.



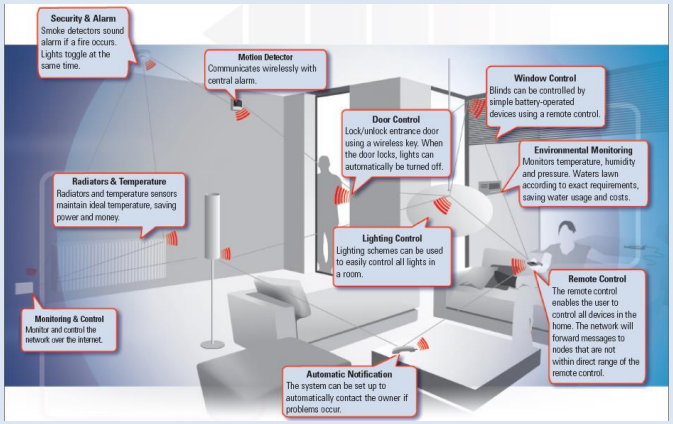
- Environment
  - Disaster detection (e.g., forest fire, flooding, etc.),
  - Ecological surveillance of an area,



- Health
  - Patient monitoring



- House automation systems
  - Smart buildings,
  - Security,

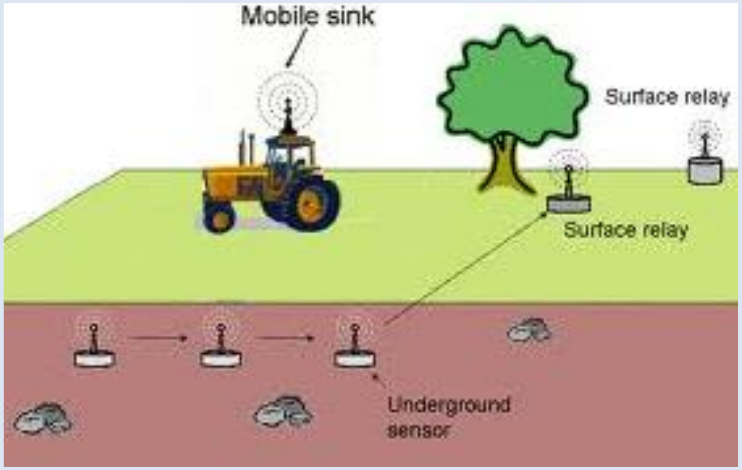


- Commercial applications
  - Industrial automation,
  - Traffic control (smart highways),
  - Building audit,

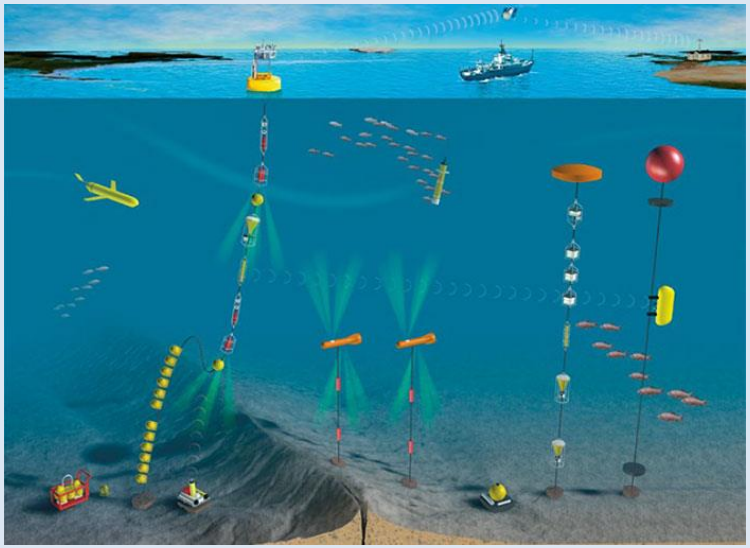


WSN Types

- Underground Wireless Sensor Networks

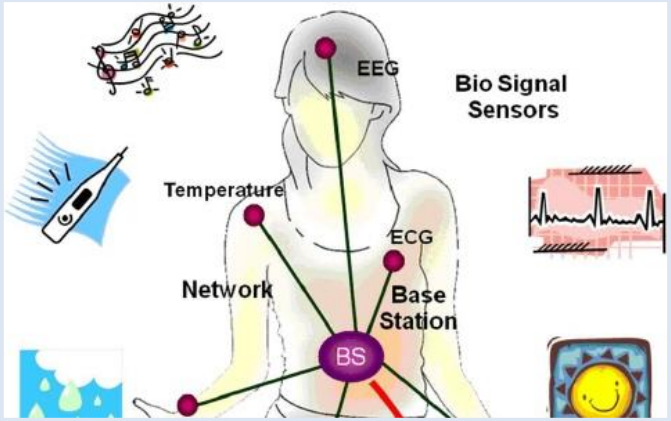


- Underwater Wireless Sensor Networks



WSN Types

- Wireless Body Sensor Networks

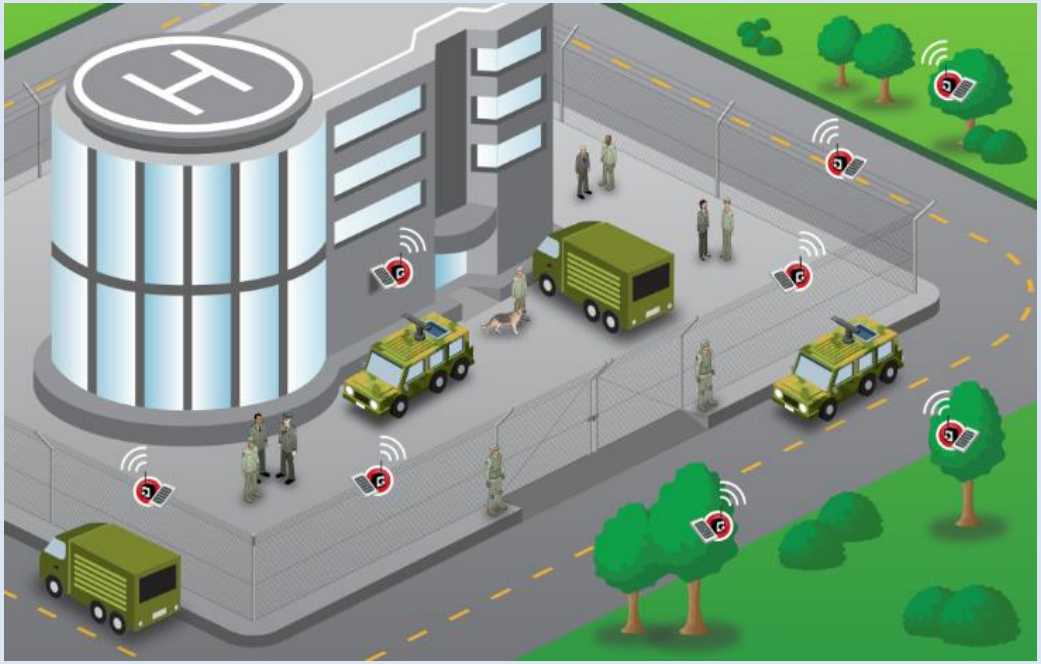


- Vehicular Area Networks

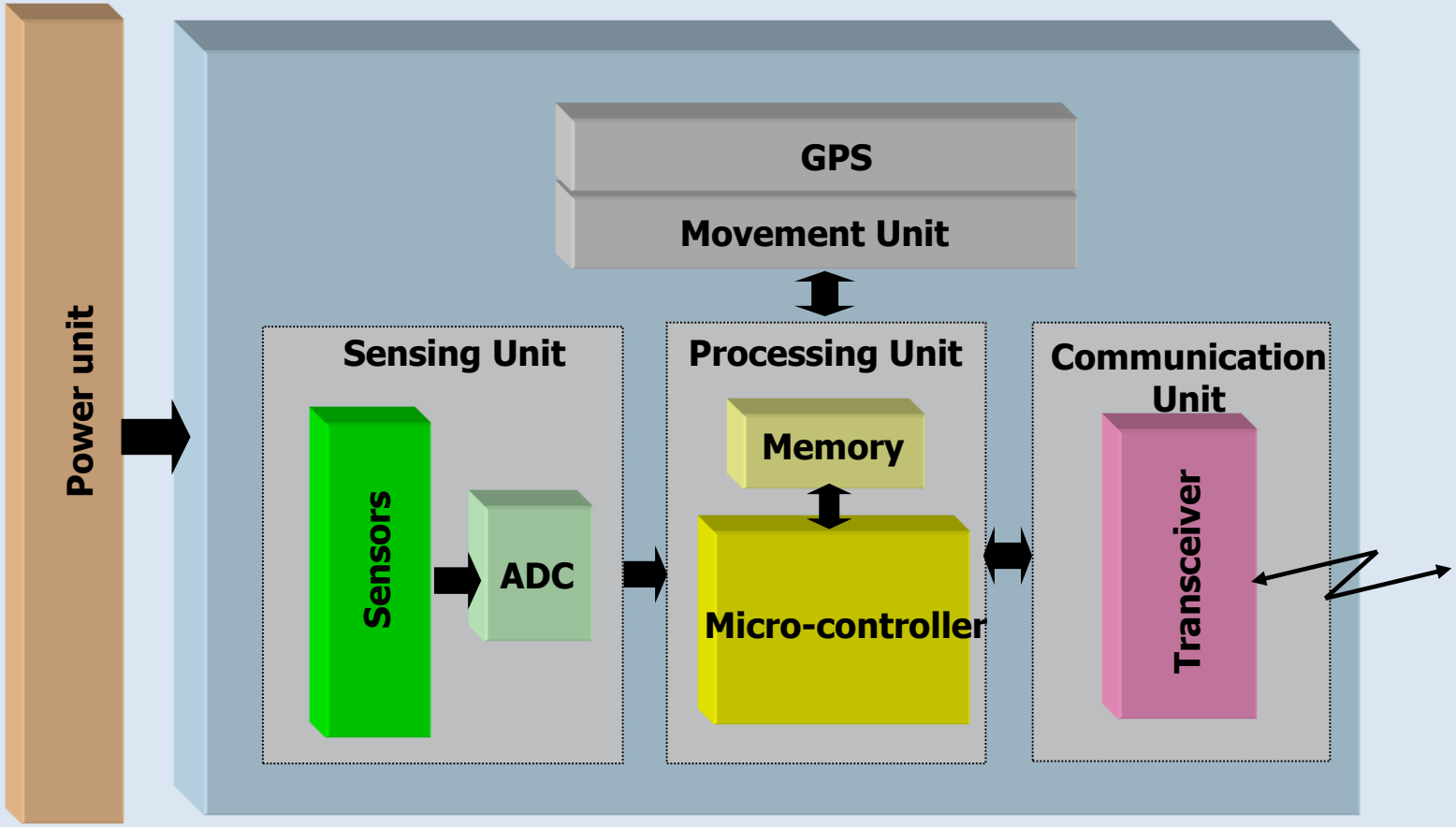


WSN Types

- Wireless Multimedia Sensor Networks



WDN Node Structure



WSN Node Types

WeC 99
"Smart Rock"



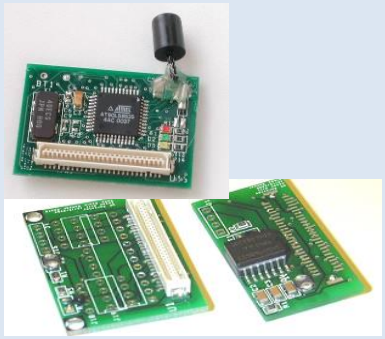
Small uC
8 kB Code memory
512 B Data memory

Simple radio
10 kbps ASK

EEPROM (32 KB)

Simple sensors

Rene 11/00



Experimental-oriented

-Sensor boards

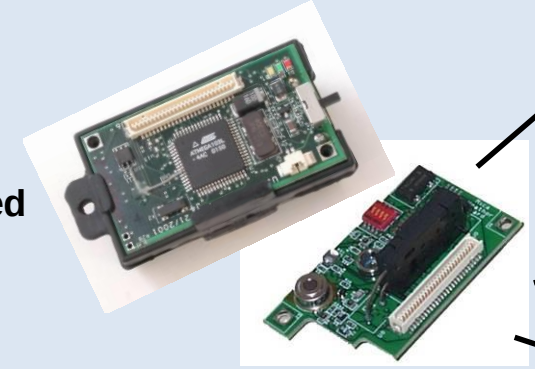
-Power boards

Dot 9/01



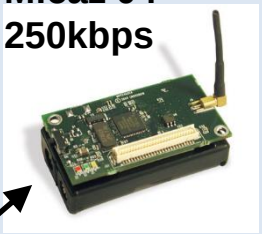
Size scale

Mica 1/02



128 kB Kod, 4 kB data
40kbps OOK/ASK radio
512 kB Flash

Micaz 04
250kbps



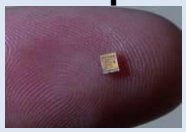
Telos 4/04
Robust, 250kbps
Easy use



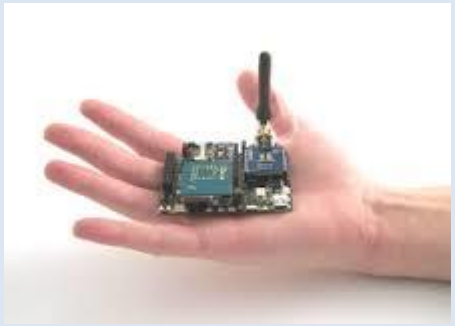


Mica2 12/02
38.4kbps radio

Spec 6/03
"node in the shape of chip"



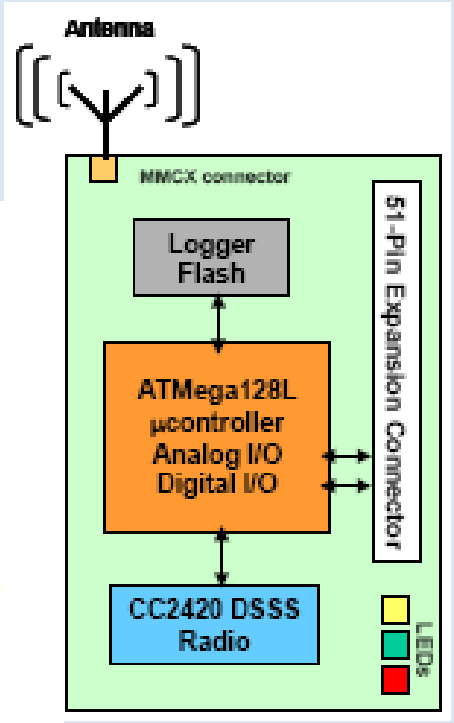
Waspmote



MICAz

- CPU:
  - 8-bit, 16 MHz, 16 MIPS ATMega128L µC
- Radio:
  - Frequency: 2400-2483.5MHz (ISM)
  - Transmission rate: 250Kbps
  - Range: 75-100m
- Memory:
  - 128 KB code memory (Flash)
  - 4 KB SRAM
  - 4 KB EEPROM
- Sensors and ADC
  - 8-channel, 10-bit ADC
  - light, sound, heat, pressure, humidity
- 3 user-configurable LED
- 2xAA battery

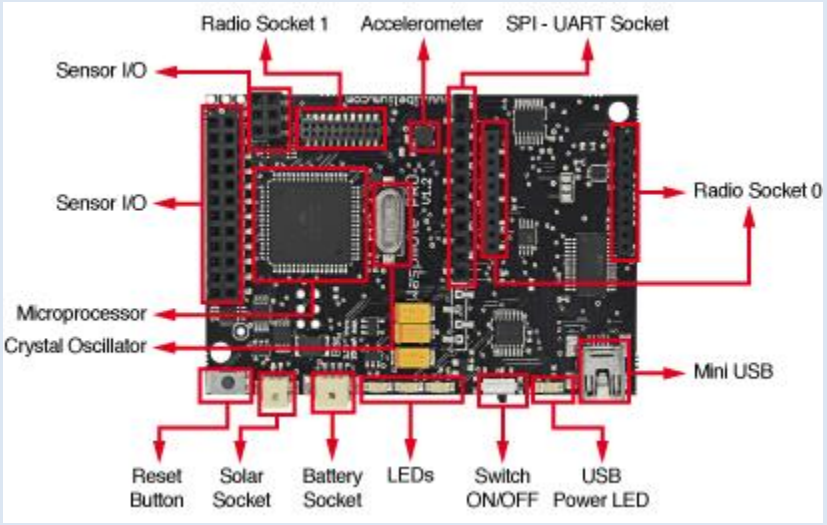




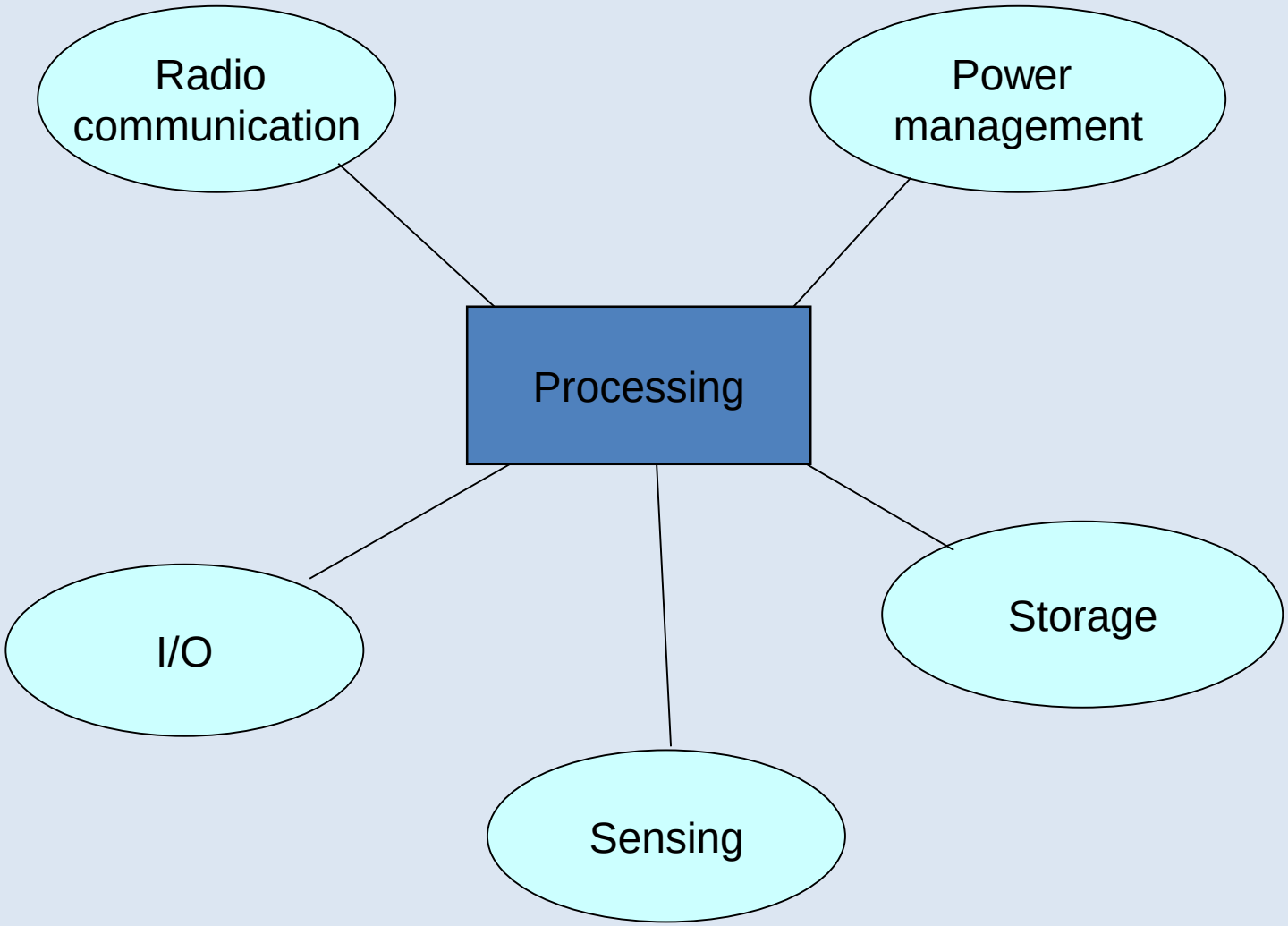
MicaZ

Waspmote

- CPU:
  - 8-bit, 14 MHz, ATMega1281 µC
- Wireless interfaces:
  - 802.15.4 / ZigBee
  - BLE (Bluetooth 4.0)
  - WiFi
  - 6LoWPAN / IPv6 Radio
  - GSM /GPRS
  - RFID/NFC
- Memory:
  - 128 KB code memory (Flash)
  - 8 KB SRAM
  - 4 KB EEPROM
  - 2 GB SD Kart
- I/O:
  - 7 Analog I/O, 8 Digital I/O, 1 I2C, 1 USB, 1 SPI
  - Basic Sensors: light, sound, heat, pressure, humidity
- 3.3v- 4.2v



WSN Logical Structure



WSN Logical Structure: Processing (MICAz)

- **Tasks:**
  - Running applications
  - Resource management
  - Peripherals control

- **Atmel AVR ATMEGA128L**
  - 16 MIPS running in 16 Mhz
  - RISC Architecture
  - 133 instructions – Mostly one-cycle instructions
  - 8 bit ALU/data path
  - 128 Kb Code memory
  - 4 Kb SRAM – Data memory
  - 4 Kb EEPROM
  - 53 programmable I/O lines
  - 3 timers, 2 UARTs, 1 SPI port
  - JTAG debugging support

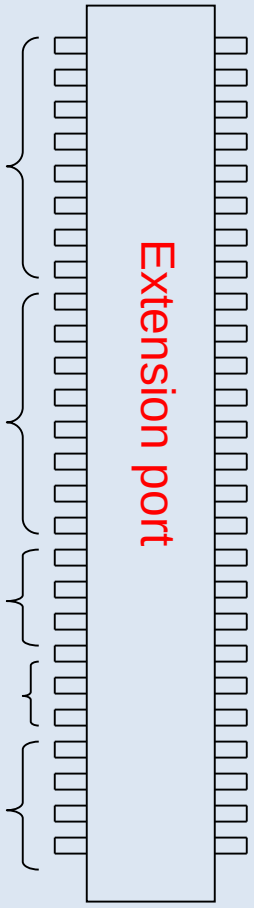


ATMEGA128

WSN Logical Structure : I/O Units (MICAz)

- **Tasks:**
  - Interface between sensor boards
  - Interface between programming boards
  - Communication between other devices

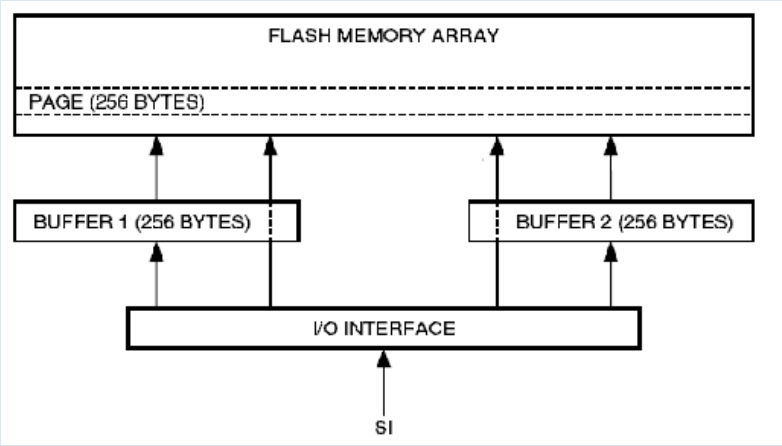
- **I/O unit consists of 51-pin extension port**
  - 8 analog line
  - 8 power management line
  - 3 PWM line
  - 2 analog comparison line
  - 4 external cut line
  - 1 I2C-line
  - 1 SPI line
  - 1 serial port
  - Microcontroller programming lines



WSN Logical Structure : Storage (MICAz)

- **Tasks:**
  - Stores sensed values
  - Temporarily stores data from network

- **Atmel AVR ATMEGA128L**
  - 4 Mb (512 kB) memory
  - 2.5V - 3.6V or 2.7V - 3.6V supply
  - Serial Peripheral Interface (SPI) compatible
  - 20 MHz maximum clock signal
  - Two 264-byte SRAM data buffer
    - Allows receiving data while programming
  - Low power consumption
    - While reading 4 mA
    - Standby 2 µA

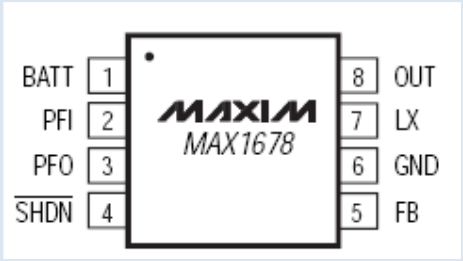


WSN Logical Structure: Power Management

- **Tasks:**
  - Regulates system power supply.

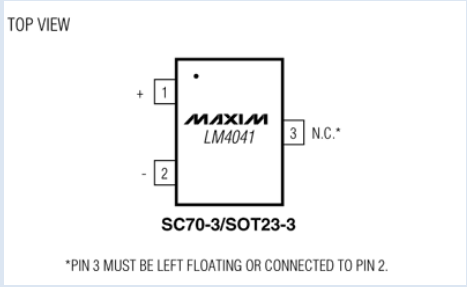
- **Maxim1678 DC-DC converter provides constant 3.0V supply**





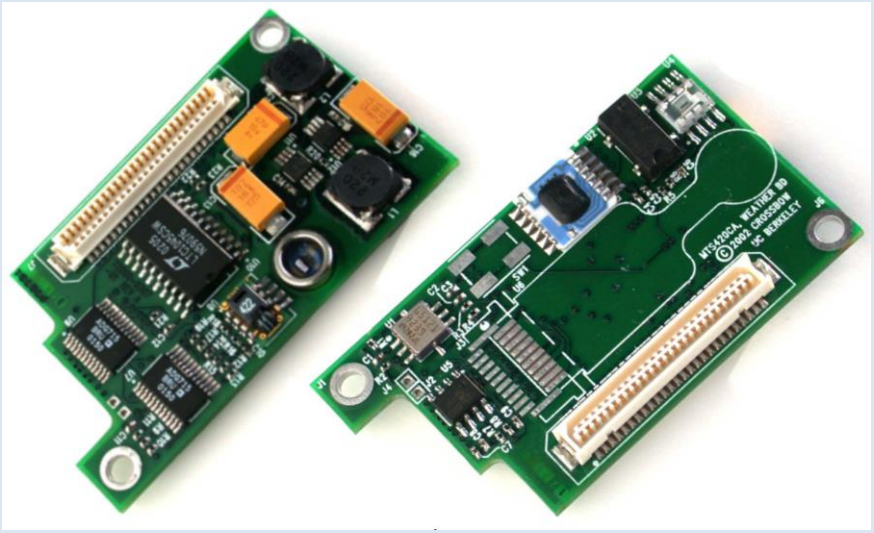
3 V

- **Batteries supply power between 3.2V and 2.0V.**
- **In 50% of Alcaline batteries, power falls below 1.2 V.**
- **Converter takes power upto 0.8V and increases it to 3.0V.**
- **3V constant power is required for radio communication.**
- **LM 4041 (voltage reference)**
  - **Regulates battery voltage.**



WSN Logical Structure : Sensing (MICAz)

- **Tasks:**
  - Senses physical attributes around
  - Translates sensed values to a form that microcontroller can understand
- **Sensor types**
  - Heat, humidity, pressure
  - Sound, light
- **Sensor-uC interface**
  - 51-Pin Connector
- **ADC**
  - 8-channel, 10-bit ADC



WSN Logical Structure : RF Communication

- **Tasks:**
  - Wireless data transmition and receiving
  - Connection to other nodes
  - Physical layer of the node.

- **Radio CC2420 (802.15.4 ZigBee)**
  - Sends data to processor in packets
  - Modulation, demodulation
  - Protocol processing
  - Synchronization
  - Coding, decoding
  - Error check and correction
  - Automatic ACK
  - 128-bit AES encryption



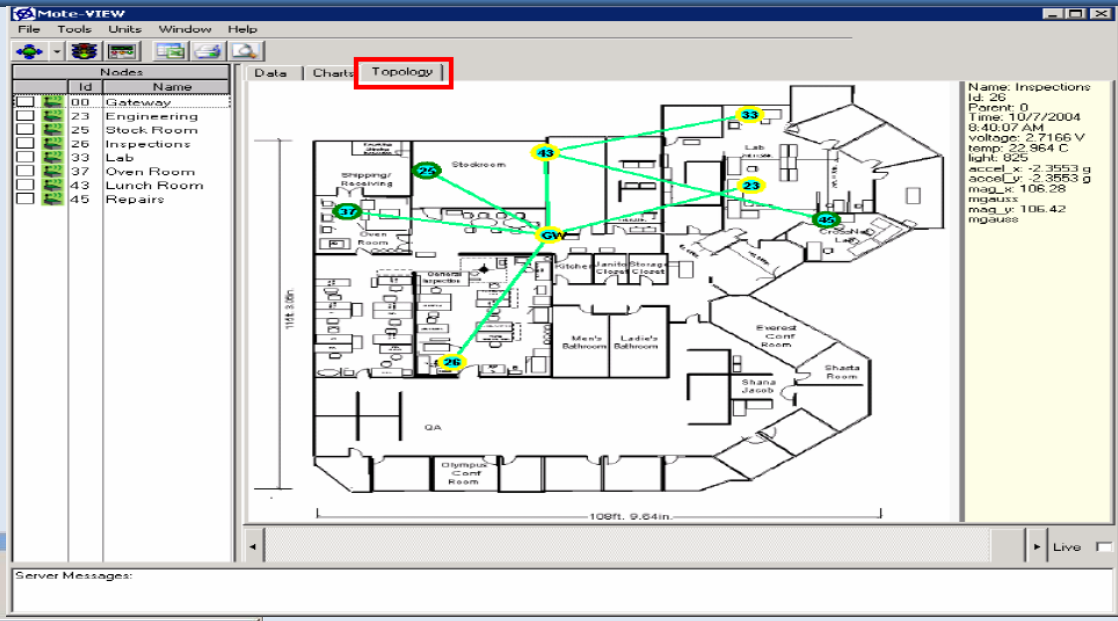
	MICA z CC2420
Radio Frequency [MHz]	2400 - 2483.5
Max data rate (kbps)	250
RX power (mA)	19,7
TX power (mA)	17,4
Low power mode (µA)	1
Modulation	DSSS-O-QPSK
Receiver sensitivity	-94 dBm
Coverage area	75 m - 100 m
Mult-channel support	Yes

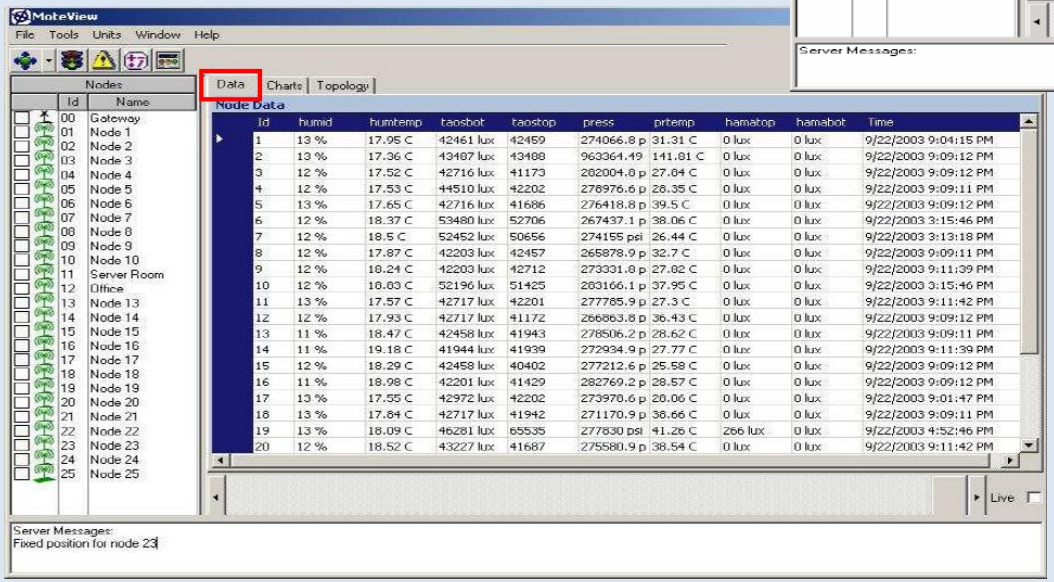
WSN Management/Application Software

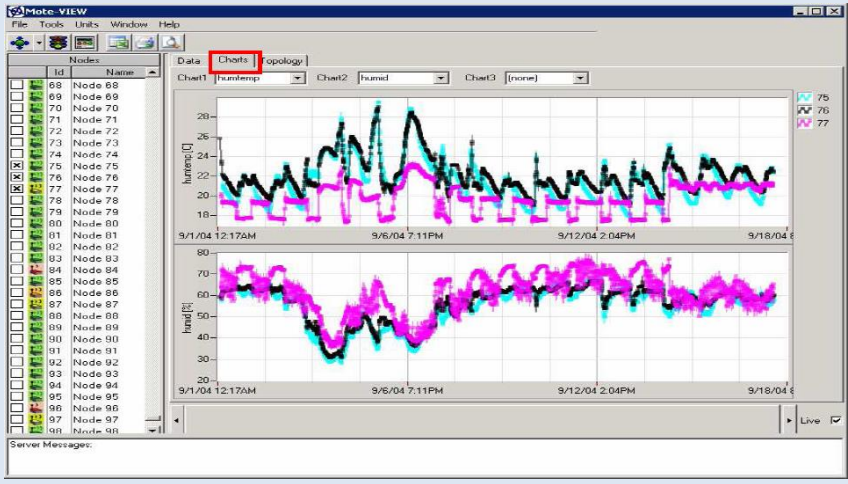
- Management tools are required for real applications.
  - Monitoring node status
    - Nodes may fail due to environmental effects..
    - Network topology may change
- Visualizing data
  - Translating data sensed by nodes to human readable form
- Reprogramming the network
- Tasks of a sensor management protocols
  - Enable/disable certain nodes
  - Management actions of the nodes
  - Query about network settings and status, re-configuring the network
  - Authentication, key sharing
  - Time synchronization between nodes
  - Data gathering and node clustering based on given rules

WSN Management/Application Software

- Example: Crossbow MoteVIEW







Limitations of WSN

- **Fabrication cost:**
  - Since thousands of nodes can be needed, it should be cheap.
- **Hardware limitations**
  - Cheap = limited resources: 8 bit, 16 Mhz uC, short range communication
- **Power consumption:**
  - Mostly unchangeable 2XAA battery
- **Environmental conditions:**
  - Harsh nature conditions: high temperature, pressure, etc.
- **Error margin (tolerance):**
  - Failure-prone due to WSN's nature: loss, malfunctioning, etc.
- **Transmission medium:**
  - Limited radio and harsh nature conditions
- **Scalability:**
  - Clustering and broad scale support

Z-Wave

- RFC G.9959: a protocol certified by ITU-T.
- A wireless communication technology designed for communications to electronic devices at home or Office, commonly used in smart building systems.
- The most important featuer of Z-Wave technology is the ability of routing for all nodes in the network. Therefore, both coverage problem and wireless data transfer problem are handled.
- All Z-Wave devices can communicate to each other by routing on mesh network and a failure on one device does not affect whole network.
- +35 million Z-Wave devices are being used. The companies that manufacture Z-Wave compatible devices: LG, Bosch, Zyxel, Honeywell, etc.

Z-Wave

- Technical Features of Z-Wave technology
  - 868 / 908 / 2400 MHz frequency band,
  - CSMA/CA medium access management,
  - 40 Kbit/s data transmission rate,
  - 232-node scalability,
  - 128 bit AES encryption,
  - 30m indoor, 100m outdoor coverage,
  - Mesh topology,
  - Fullduplex data transfer,
  - İndependent or cluster-wise operations by Z-Wave devices



ANT/ANT+

- ANT is short-range ultra-low power consumption, low-data transmission rate wireless technology designed for sensor networks, health and sport (fitness) applications.
- ANT, developed by Dynastream, was acquired by Garmin, a bicycle and sport equipment producer
- Since it is a registered product, it couldn't become a widely used technology.
- ANT+ technology is expected to become more wide-spread in smart phones. Ex: Samsung S7, S6, S5, Sony Xperia XZ, Z5
- ANT nodes/devices can work as master and slave.
- Technical features
  - 2.4 GHz ISM band,
  - TDMA medium access management,
  - 232-node scalability,
  - 1-30 m coverage area,
  - 40 Kbit/s
  - Low power consumption



References

- http://www.z-wave.com/
- https://www.youtube.com/watch?v=e_Nmj9KtM34
- https://www.thisisant.com/consumer/ant-101/what-is-ant/